

Data Science Environment Setup

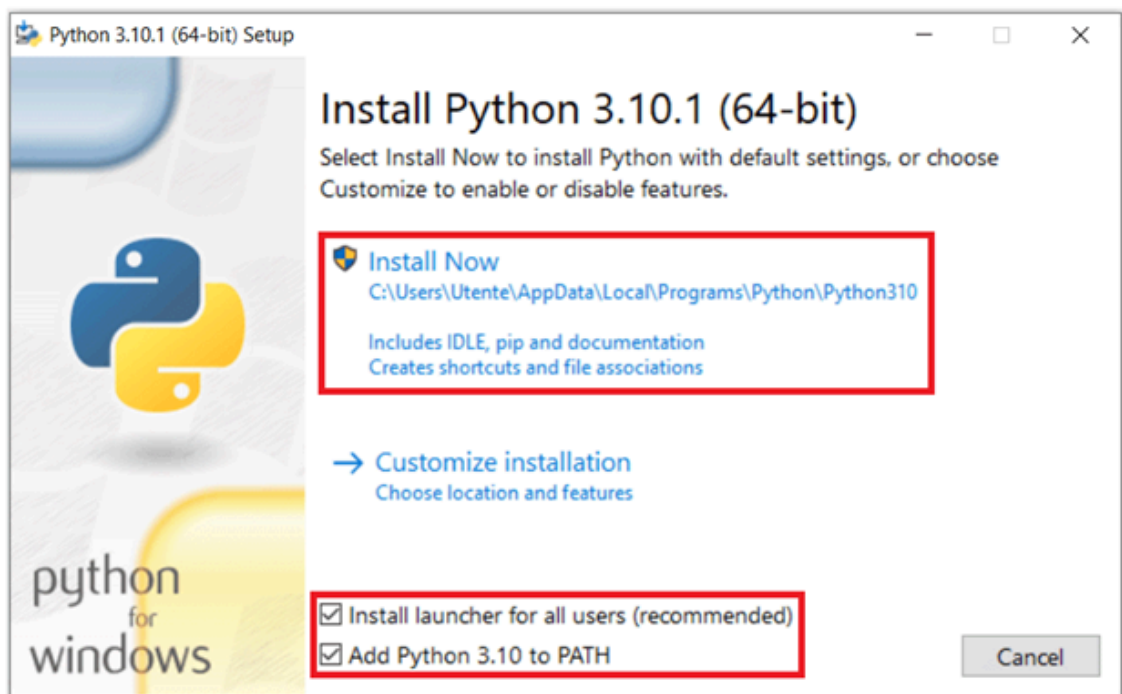
This document walks through installing Python, Visual Studio Code, and a Jupyter-friendly workflow.

1. Install Python and add it to PATH

Why this matters: *PATH* means your computer can find the `python` program from any folder. If Python is not on PATH, later steps (terminal commands, VS Code) will fail with *command not found*.

Windows

1. In a browser, open <https://www.python.org/downloads/>
2. Download the latest **Python 3.x Windows installer** (64-bit is typical).
3. Run the installer. On the first screen, **check Add python.exe to PATH** (also called *Add Python to environment variables* on some versions). **Do not skip this.**
4. Click **Install Now** and wait until it finishes. Close the installer.



macOS

1. Open <https://www.python.org/downloads/> and download the **macOS** installer, **or** if you use Homebrew: in Terminal run `brew install python`.

Check that it worked

Option A — in this notebook: Run the **next code cell** (Shift+Enter). You should see a **version line** and a **full path** ending in `python` or `python3`.

Option B — in a terminal: Open **Command Prompt** (Windows) or **Terminal** (macOS) and run:

- **Windows:** `python --version` or `py --version`
- **macOS:** `python3 --version`

You should see something like `Python 3.12.x`. If you see *command not found* or *not recognized*, Python is not on PATH—on Windows, run the installer again and enable **Add python.exe to PATH**; on macOS, ensure you are using `python3` or fix your PATH with official Python/macOS help.

2. Install Visual Studio Code (VS Code)

What VS Code is: A free code editor where you will open folders, run a terminal, edit `.py` files, and open **Jupyter** notebooks (`.ipynb`).

1. Open <https://code.visualstudio.com/>
2. Download **VS Code** for **Windows** or **macOS**.
3. Run the installer; accept the defaults unless your school tells you otherwise.
4. Start **VS Code** from the Start menu (Windows) or Applications (macOS).

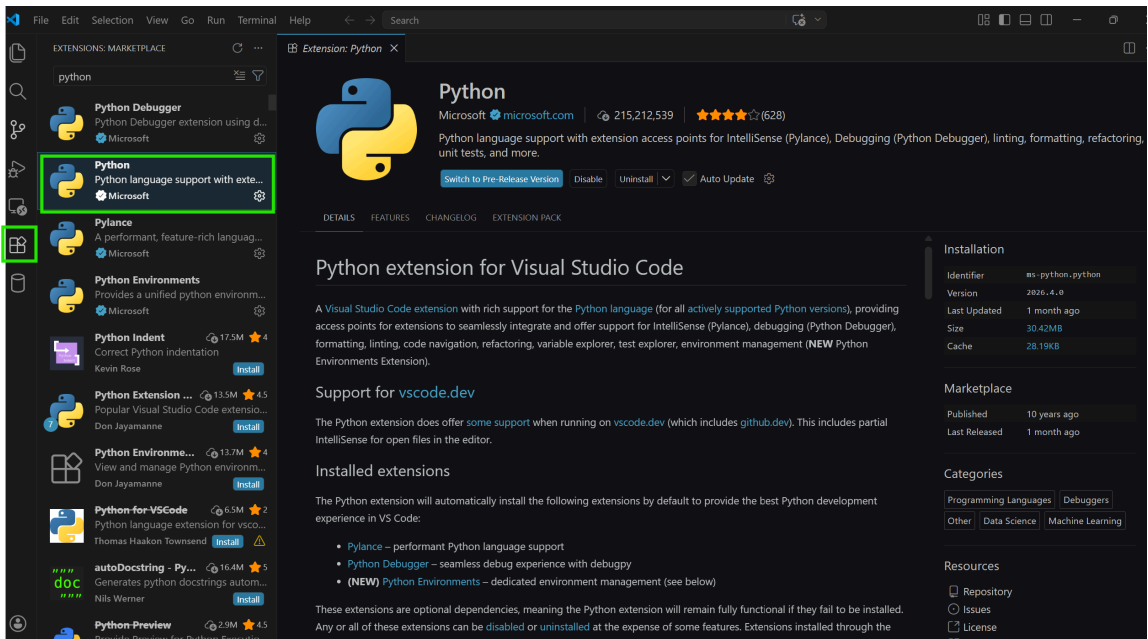
You are done with this step when: VS Code opens and you see the welcome screen or an empty window.

3. Install the Python and Jupyter extensions in VS Code

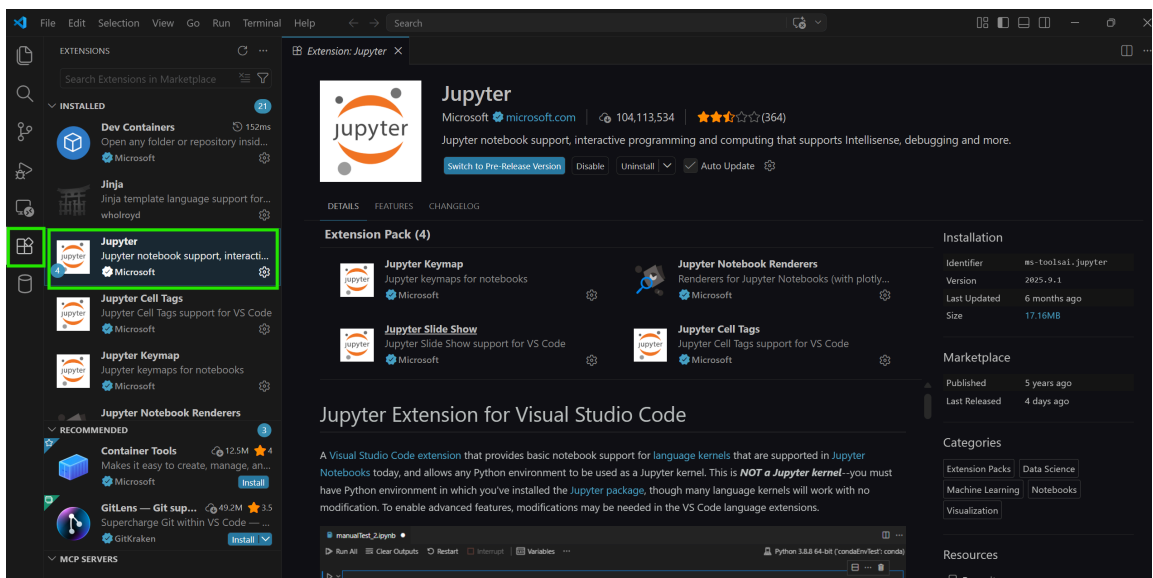
What extensions are: Small add-ons so VS Code understands Python and can run **Jupyter** notebooks.

1. In VS Code, open **Extensions**: press **Ctrl+Shift+X** (Windows) or **Cmd+Shift+X** (macOS).
2. Search for **Python**. Install the one published by **Microsoft** (name: *Python*).
3. Search for **Jupyter**. Install the one published by **Microsoft** (name: *Jupyter*).

Python extension



Jupyter extension



If that image file is missing, either add your own `jupyter_extension.png` next to this notebook or change the path above to match your screenshot filename (some courses reuse `python_extension.png` until a second image is added).

You are done when: Both extensions show **Installed** and VS Code does not ask you to reload endlessly (reload once if it asks).

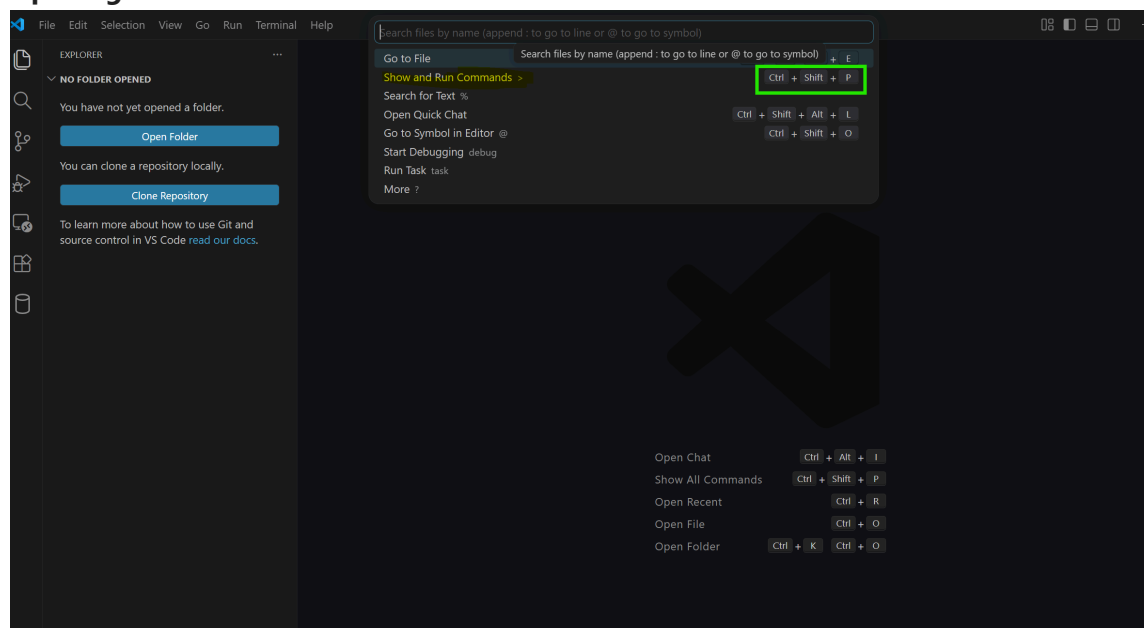
4. Windows: set Command Prompt as default terminal

1. **Settings** (`Ctrl+,`) → search **default profile windows**.
2. **Terminal** > **Integrated** > **Default Profile: Windows** → **Command Prompt**.

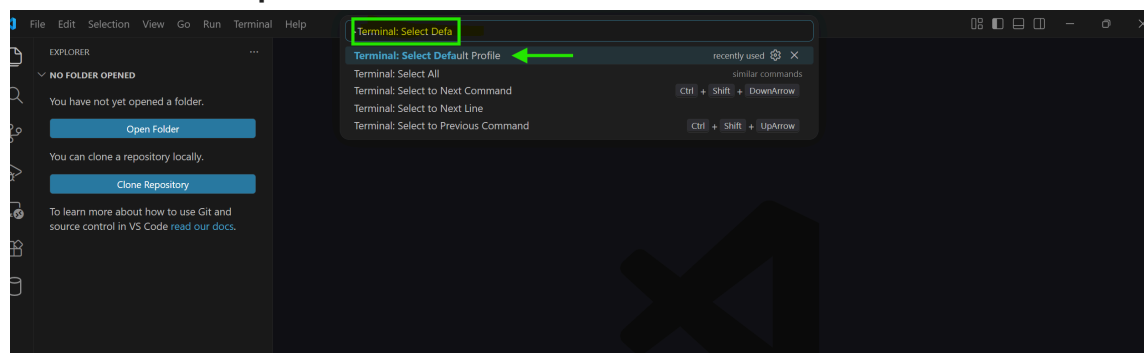
Or in **User Settings (JSON)**:

"terminal.integrated.defaultProfile.windows": "Command Prompt"

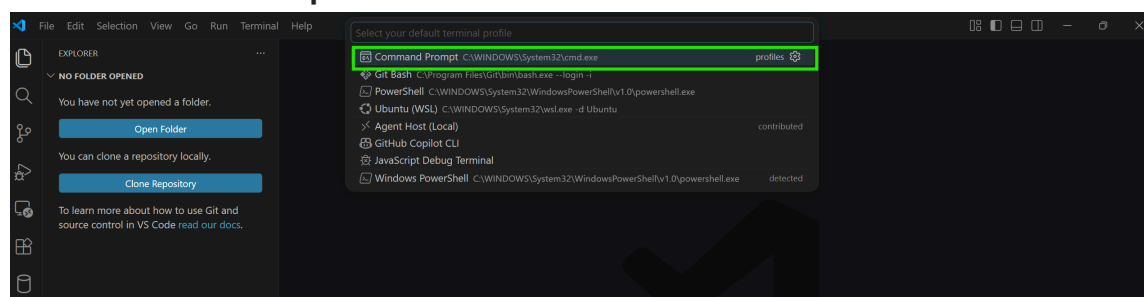
Opening Command Palette



Default Terminal Option

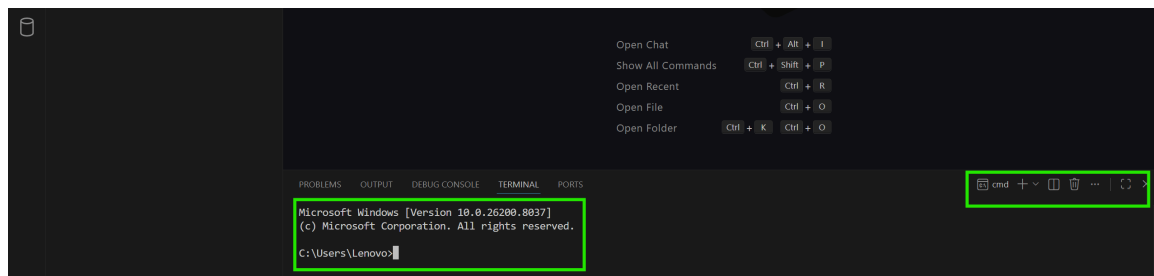


Select Command Prompt



On macOS you can leave the default shell or set **Terminal > Integrated > Default Profile: Osx** to your preference.

Open **Terminal** → **New Terminal** (default shortcut is **Ctrl** plus the **backtick(~)** key, above Tab). The integrated terminal should start as **Command Prompt**.



5. Open a new project in VS Code

File → **Open Folder...** — create or pick a folder (e.g. `ds_cl_code`).

6. Virtual environment (Windows and macOS)

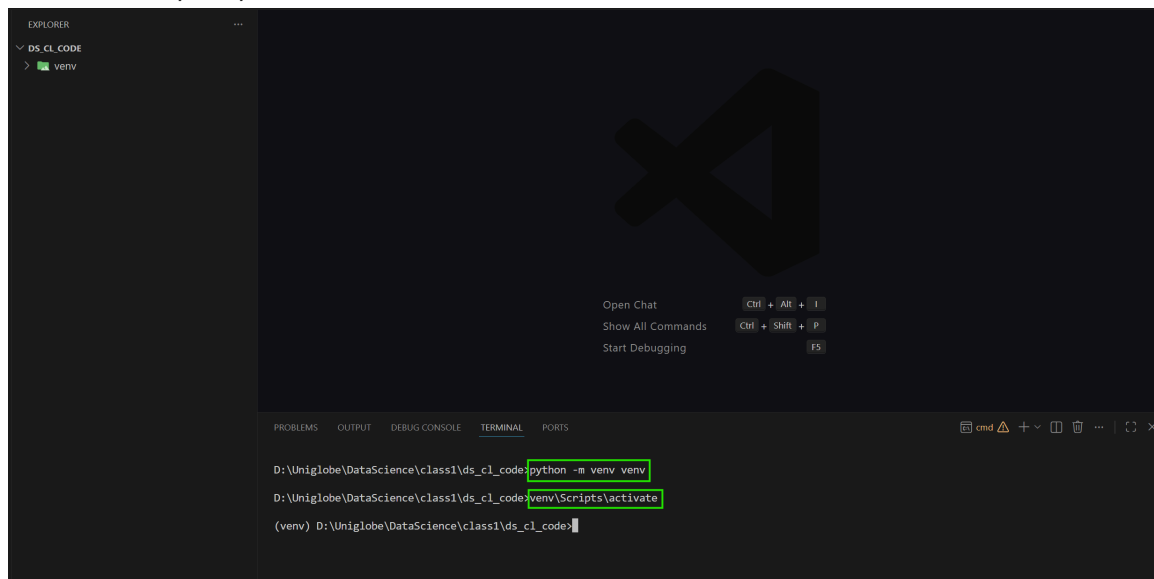
Create the venv **from a terminal** in your project folder (VS Code terminal is fine).

Windows (Command Prompt)

```
python -m venv venv
venv\Scripts\activate
```

macOS

```
python3 -m venv venv
source venv/bin/activate
```



Select interpreter in VS Code

Ctrl+Shift+P / Cmd+Shift+P → **Python: Select Interpreter** → choose
venv\Scripts\python.exe (Windows) or venv/bin/python3 (macOS).

7. Install required libraries (terminal or notebook)

With `venv` activated and selected as the interpreter, in a **terminal**:

```
pip install pandas matplotlib seaborn
```

Install Start

```
(venv) D:\Uniglobe\DataScience\class1\ds_cl_code>pip install pandas matplotlib seaborn
Collecting pandas
  Obtaining dependency information for pandas from https://files.pythonhosted.org/packages/7b/8b/721a9cff6fa6a91b162eb51019c6243b82b3226c71bb6c8e
f4a9bd65cbc6/pandas-3.0.2-cp312-cp312-win_amd64.whl.metadata
  Using cached pandas-3.0.2-cp312-cp312-win_amd64.whl.metadata (19 kB)
Collecting matplotlib
  Obtaining dependency information for matplotlib from https://files.pythonhosted.org/packages/04/a1/4571fc46e7702de8d0c2dc54ad1b2f8e29328dea3ee9
0831181f7353d93c/matplotlib-3.10.9-cp312-cp312-win_amd64.whl.metadata
  Downloading matplotlib-3.10.9-cp312-cp312-win_amd64.whl.metadata (52 kB)
52.8/52.8 kB 687.0 kB/s eta 0:00:00
Collecting seaborn
```

Install End

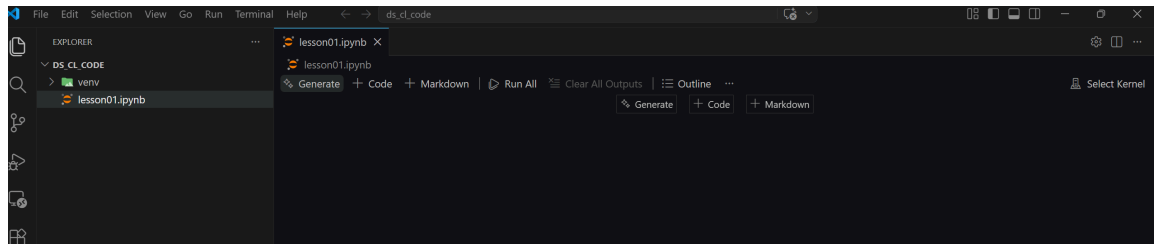
```
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Installing collected packages: tzdata, six, pyparsing, pillow, packaging, numpy, kiwisolver, fonttools, cycler, python-dateutil, contourpy, pandas, matplotlib, seaborn
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.62.1 kiwisolver-1.5.0 matplotlib-3.10.9 numpy-2.4.4 packaging-26.2 pandas-3.0.2 pillow-12.2.0 pyparsing-3.3.2 python-dateutil-2.9.0.post0 seaborn-0.13.2 six-1.17.0 tzdata-2026.2

[notice] A new release of pip is available: 23.2.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

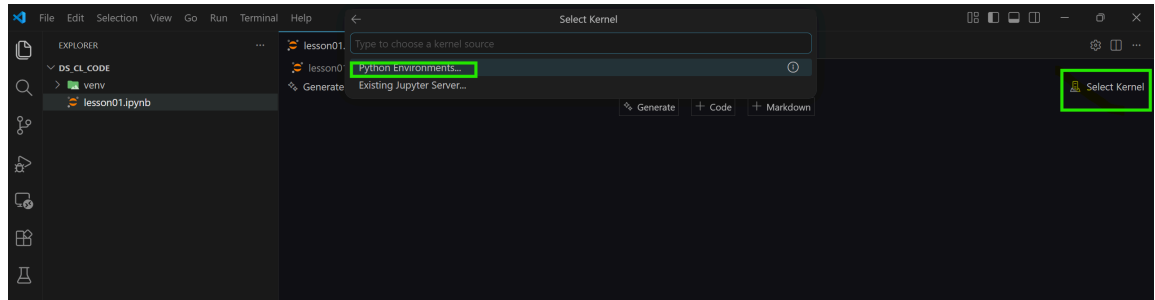
(venv) D:\Uniglobe\DataScience\class1\ds_cl_code>
```

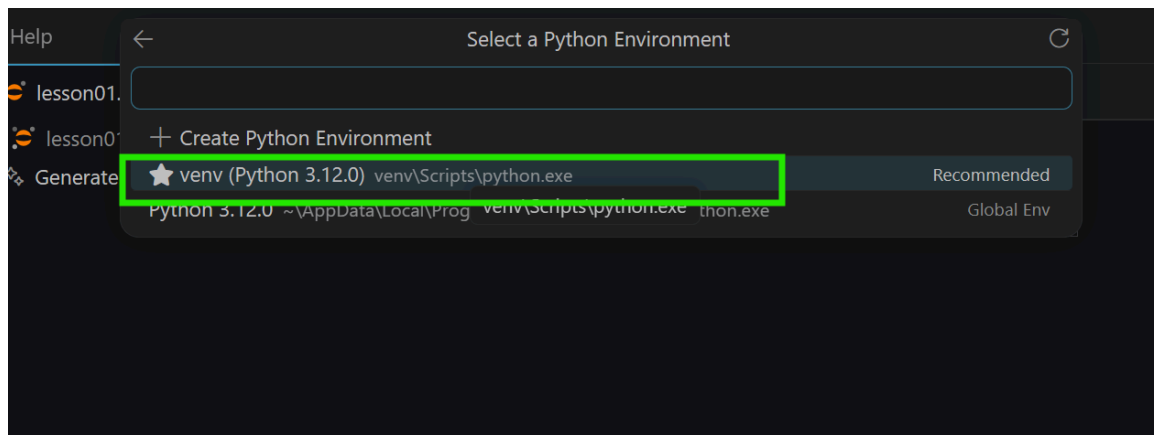
8. Create and use this notebook

Explorer → right-click folder → **New File** → name it e.g. `lesson01.ipynb`.



Select Kernel (top right) → **Python (venv)** or your venv display name.

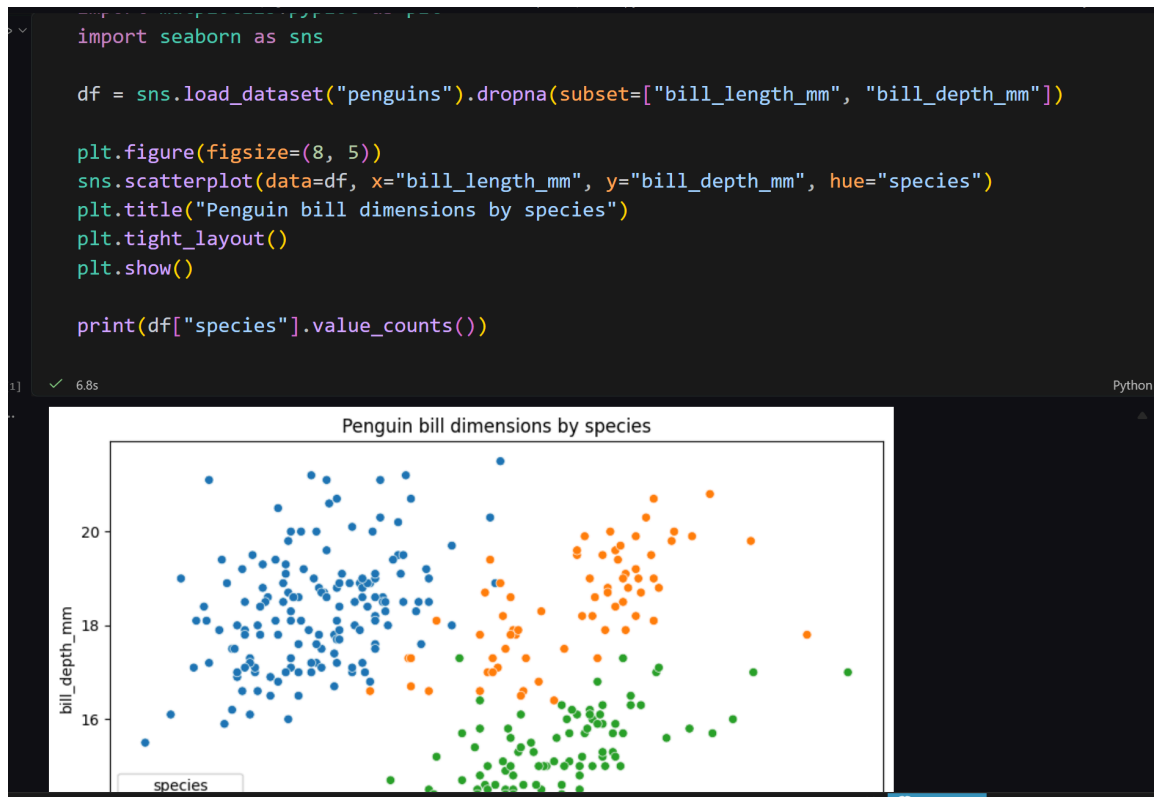




10. Example code

Run the next cell. You should see a plot and value counts.

If `sns.load_dataset("penguins")` fails (network/cache), run the **fallback** cell after it.



```
In [ ]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = sns.load_dataset("penguins").dropna(subset=["bill_length_mm", "bill_depth_mm"])

plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x="bill_length_mm", y="bill_depth_mm", hue="species")
plt.title("Penguin bill dimensions by species")
```

```
plt.tight_layout()
plt.show()

print(df["species"].value_counts())
```

Quick checklist

Step	Done
1	Python on PATH; <code>python</code> / <code>python3</code> works
2	VS Code installed
3	Python + Jupyter extensions
4	(Windows) Default terminal = Command Prompt
5	Project folder opened
6	<code>venv</code> created, activated, interpreter selected
7	Jupyter / ipykernel (and optional kernel name)
8	Notebook created; correct kernel
9	pandas, matplotlib, seaborn installed
10	Example cell runs

Tip: Use **Shift+Enter** to run a cell and move to the next.